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ENGINEERING



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Sreejith Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	Blessy S	Register Number	192421053
Class / Section	A	Slot	B
Course Code	CSA1512	Course Title	Cloud computing and big data analytics
Capstone Title	Cloud security		
Submission Date	13.7.26		
Faculty Name	Dr.Rajesh Babu	Employee ID	
Department			

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	blessy

Date	13.7.26
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Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-COAssessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

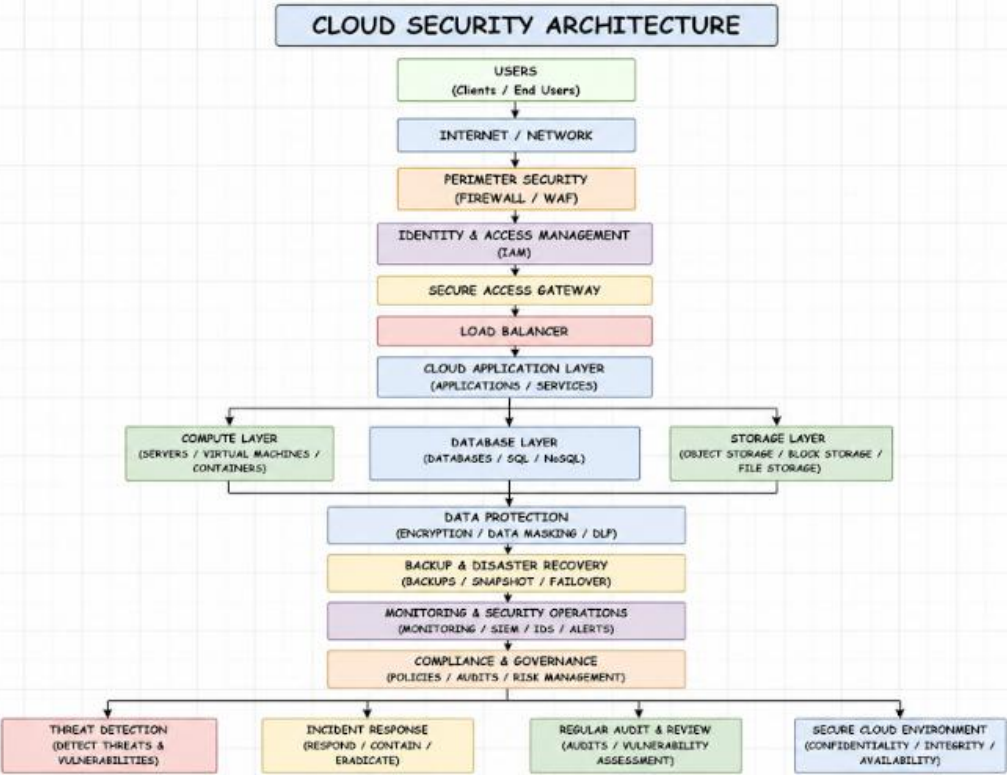
Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.

Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space.



Brief explanation of your concept map: explain the most important nodes, link labels, hierarchy, and capstone connection.

This concept map shows the Cloud Security Architecture in a hierarchical flow from users to cloud resources. Users access cloud services through the internet, where security layers such as Firewall, IAM, Secure Access Gateway, and Load Balancer protect the system. The Cloud Application Layer processes requests, while the Compute, Database, and Storage Layers manage applications and data. Data Protection, Backup & Disaster Recovery, and Monitoring ensure security, availability, and reliability. The link labels such as accesses, authenticates, protects, processes, stores, and monitors explain the relationships between nodes. For my E-commerce Website capstone, this architecture provides secure user authentication, cloud data storage, application hosting, and scalable resources to handle increasing customer traffic.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

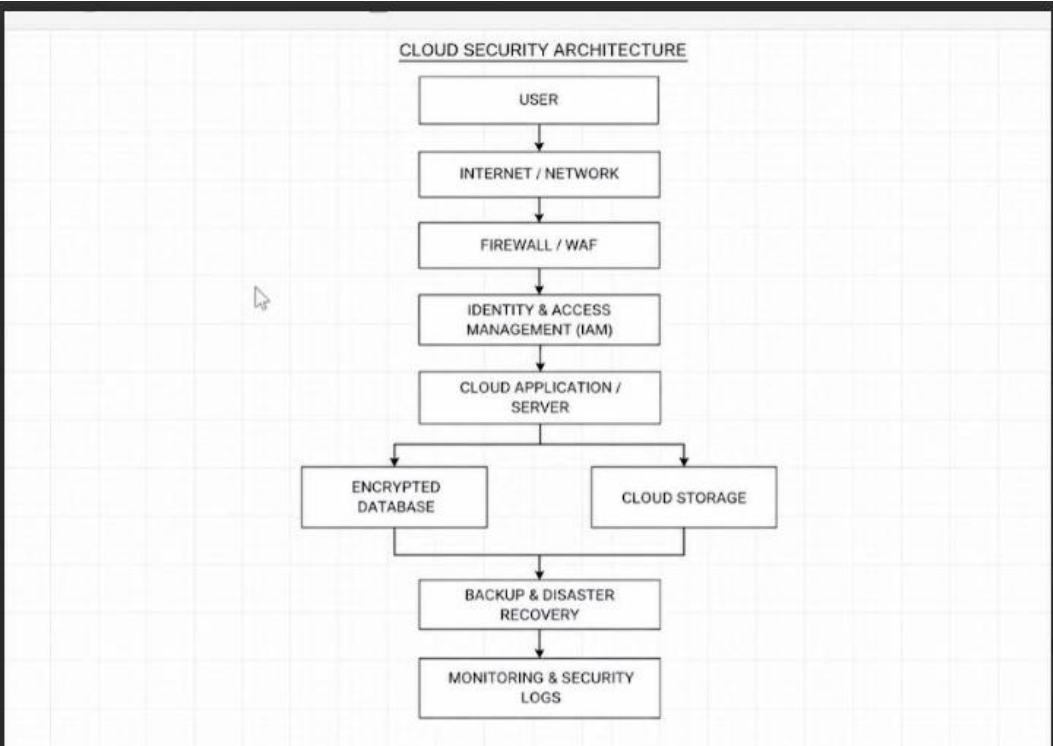
Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS	Infrastructure Security (Servers, Network)	Provide virtual machines, storage, networking, virtualization	Configure OS, firewall, security settings, access control	VM configuration screenshot
PaaS	Secure Application Development	Manage runtime, middleware, platform, scaling	Develop and deploy applications, implement authentication	Application deployment screenshot
SaaS	Security Monitoring Dashboard	Maintain software, updates, backups, availability	Use dashboard, monitor alerts, view reports	Dashboard/Monitoring screenshot
XaaS / Specialized Service	Identity & Access Management (IAM)	Provide IAM, authentication, MFA, access control services	Create users, assign roles, manage permissions	IAM configuration screenshot

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI	Web Browser / React	User requests, pages, input data	HTTPS	Secure access, responsive UI, CDN for scaling
API / backend logic	Node.js / Express	Request processing, business logic	REST API	Authentication, load balancing, auto scaling
Authentication and identity	Firebase Authentication / IAM	User credentials, tokens, roles	OAuth / JWT API	MFA, role-based access control
Database / structured records	MySQL / MongoDB	User data, security logs, application data	Database API	Encryption at rest, access control, backups
File / object storage	AWS S3 / Azure Blob Storage	Documents, backups, files	Storage API	Encryption, secure backup, scalable storage
Monitoring / logs / usage metrics	AWS CloudWatch / Azure Monitor	Logs, metrics, alerts	Monitoring API	Real-time monitoring, auto alerts, scalable
Analytics / reporting output	Power BI / Grafana	Security reports, analytics data	REST API	Scalable dashboards, visual analytics

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

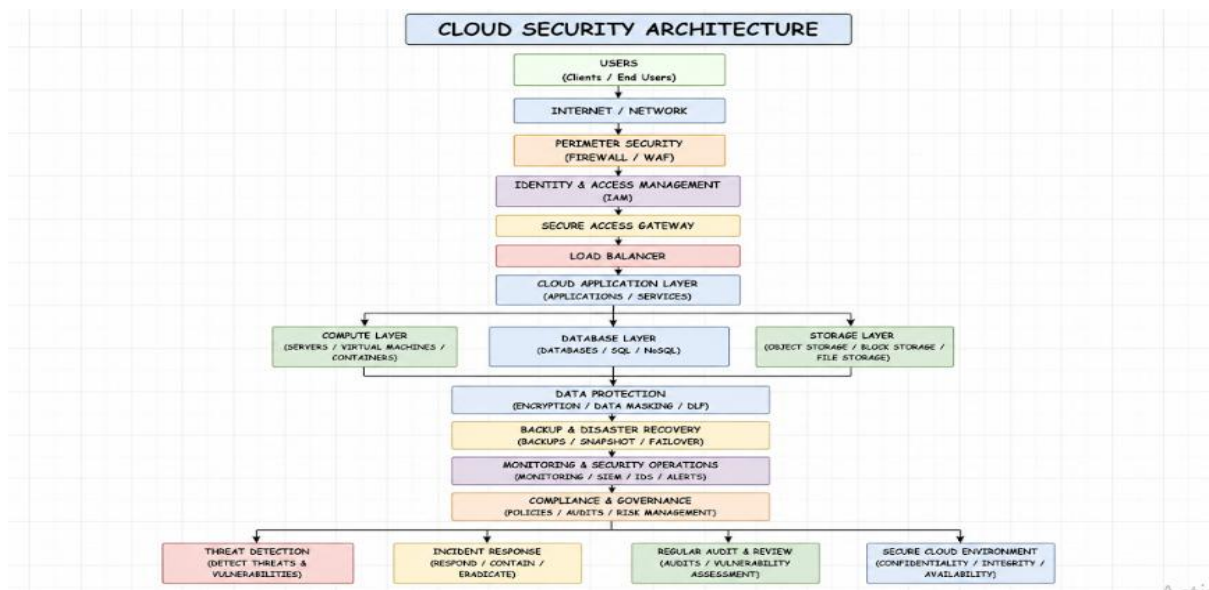


Student Drawing Requirement

Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	
Frontend / app screen	
API endpoint and method	
Backend processing rule	
Cloud database / storage operation	
Response, alert, report, or dashboard output	
Security, scalability, and elasticity labels added in diagram	

Paste your redrawn capstone cloud-flow diagram here.



Brief explanation: explain one complete request-response path from user action to cloud response.

A user first accesses the cloud application through a web browser or mobile app. The request is sent over the internet and passes through the Firewall and Identity and Access Management (IAM) for authentication and security verification. After successful authentication, the request reaches the Cloud Application Layer, where it is processed. The application retrieves or stores data in the Database or Cloud Storage through secure APIs. The cloud server then sends the response back to the user. Throughout this process, data encryption, monitoring, backups, and security controls protect the system, while scalability and elasticity ensure the application can handle increasing user requests efficiently.

Task 4 - Individual Reflection

Answer the following from your own perspective. Avoid team-common wording.

1. What is the strongest cloud concept represented in your capstone project?

The strongest cloud concept represented in my capstone project is Cloud Security. The project focuses on protecting cloud-based applications, data, and services from unauthorized access and cyber threats. It includes multiple security layers such as Identity and Access Management (IAM), firewalls, secure access gateways, data encryption, monitoring, and backup mechanisms. These components work together to maintain the confidentiality, integrity, and availability of cloud resources. The project also demonstrates how cloud providers and users share security responsibilities to create a reliable and secure cloud environment. By implementing these security measures, the cloud system can safely store sensitive information, authenticate users, detect threats, and ensure business continuity even during failures or attacks.

2. Where will elasticity or scalability become important in your capstone product?

Elasticity and scalability become important when the number of users, applications, or security requests increases. During peak usage, such as when many users access the cloud application simultaneously, the cloud platform can automatically allocate additional computing resources such as virtual machines, storage, and network bandwidth. This prevents system slowdowns and ensures continuous service availability. When the workload decreases, the extra resources are automatically released, reducing operational costs. Scalability also helps the system manage growing amounts of data, security logs, and monitoring activities without affecting performance. These cloud features make the security architecture efficient, cost-effective, and capable of handling future growth.

3. Which API or web-service idea is most relevant to your project and why?

The REST API (Representational State Transfer) is the most relevant web service for my project because it provides secure and efficient communication between cloud applications, databases, authentication services, and monitoring systems. REST APIs use standard HTTP methods such as GET, POST, PUT, and DELETE to exchange data between different cloud components. They are lightweight, scalable, and easy to integrate with modern web and mobile applications. In my cloud security architecture, REST APIs are used for user authentication, accessing cloud resources, retrieving security logs, storing data, and communicating with cloud services. Combined with HTTPS and authentication tokens, REST APIs help ensure secure data transmission and reliable communication.

4. What did you personally understand better after preparing this concept map and matrix?

After preparing this concept map and service model matrix, I gained a much better understanding of how cloud computing components are connected to build a secure cloud environment. I learned the differences between IaaS, PaaS, SaaS, and XaaS, as well as the responsibilities of both cloud providers and users. I also understood how virtualization, APIs, cloud storage, authentication, monitoring, and data protection work together within cloud architecture. Creating the concept map helped me visualize the relationship between different cloud concepts, while the service model matrix improved my understanding of selecting appropriate cloud services for different requirements. Overall, this activity strengthened my knowledge of cloud security architecture and its role in developing secure, scalable, and reliable cloud-based applications.

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.
Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or generic non-technical content.
Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	